

Fundamentals 1: Biocompatibility, atomic building blocks, structure of materials

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Biocompatibility

- A material's compatibility with a host to elicit an appropriate response: non-deleterious and beneficial
 - Does not necessarily mean *inert*
- Differs from safety, which is determined by the risk assessment of a material

Unwanted reactions to materials:

- Toxicity
- Contact dermatitis
- Paresthesia
- Lichenoid reactions
- Anaphylactoid reactions
- Urticaria
- Intolerance

Unwanted interactions between biological environment and material:

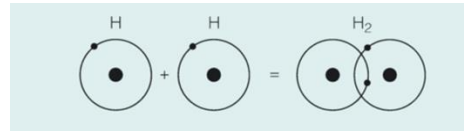
- Corrosion (i.e. amalgam and saliva – discoloration, margin breakdown) - releases corrosive products, potentially eliciting reactions

Atomic building blocks

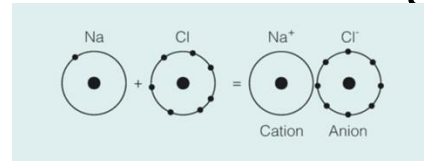
- Atoms want to be in the lowest energy state, which is achieved by having a complete outermost electron orbital
 - Usually involves atoms being in close proximity for stability
- This can be achieved by bonding to other atoms in proximity

Types of atomic bonds (listed from highest to lowest strength)

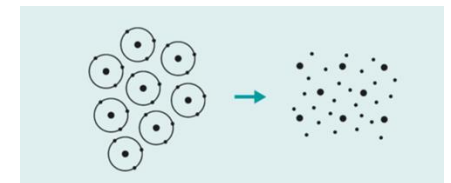
- Covalent bonding - atoms share their electrons to complete their outermost electron orbital; directional bond as electrons become localized to overlapping region



- Ionic bonding - an atom transfers its valence electron to a recipient ion, and both atoms achieve an inert structure; oppositely charged ions are attracted to each other, approximating them and reducing total energy

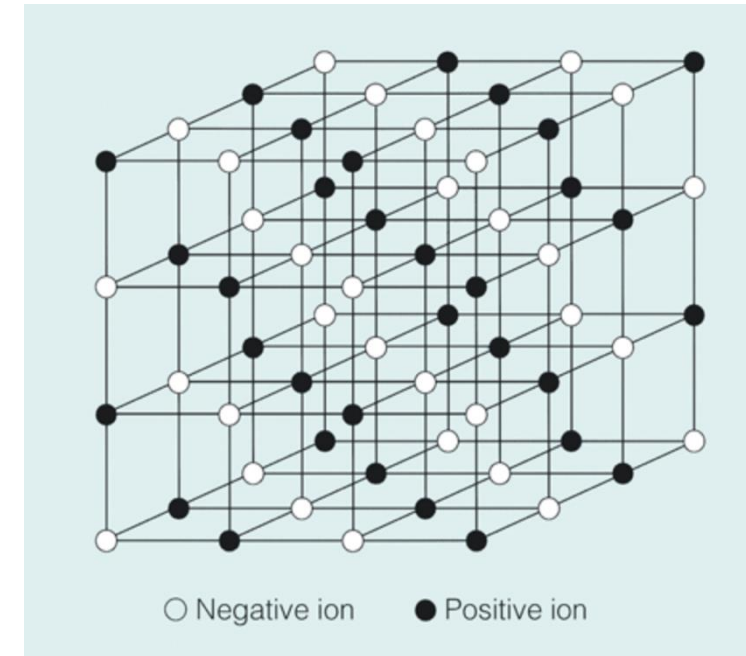


- Metallic bonding - aggregate of metal atoms, with free-flowing electrons between all electron orbitals in close proximity, achieving lower energy



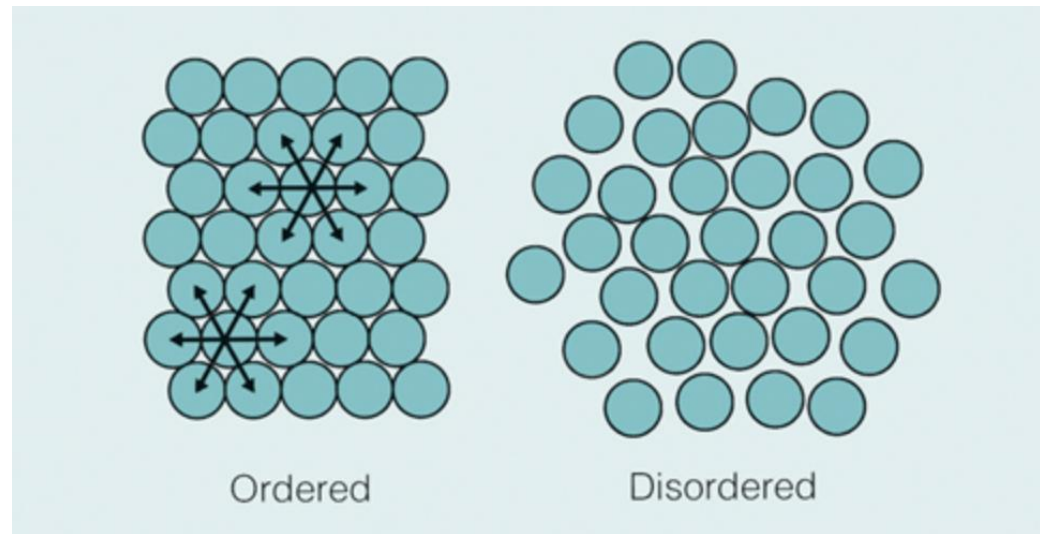
Formation of solids

- Ionic solids - ions surrounded by non-directional electrostatic fields to form a regular, 3D network (i.e. ceramics, glasses)
- Metallic solids - lacks strong electrostatic attraction between individual atoms, forming a cloud of electrons (i.e. metals)
- Covalent solids - rare due to directionality of bonding (i.e. Diamonds)
- Molecular solids - chains of hydrocarbons (C_nH_{2n+2}) bonded to create polymers



Structural arrangement

- Crystalline - configuration with minimum energy; atoms are arranged and packed closely
- Amorphous – atoms are in an asymmetrical, disordered fashion



Structure of ceramics

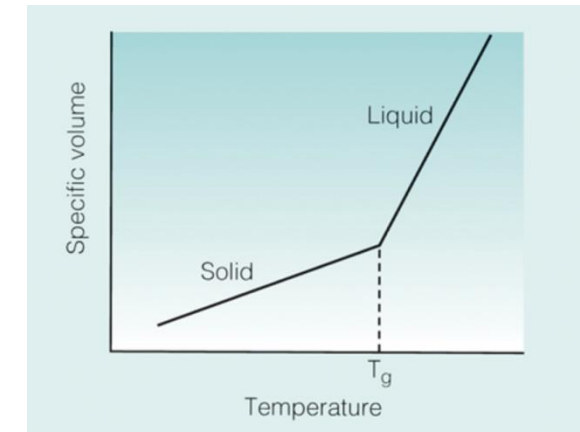
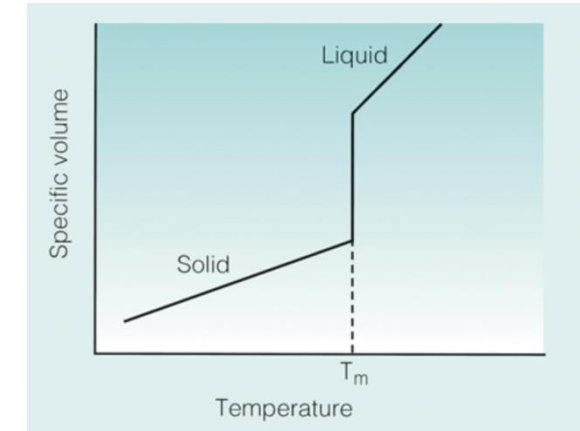
- Compounds of metallic and nonmetallic substances
- Can be crystalline or amorphous (glass) phases
 - Intermediate class is termed 'glass ceramics' - both phases
 - For dentistry, can combine components of both phases to create favorable effects (esthetics, function)

Silica - the basis of many ceramics

- SiO_2 - crystalline material with many forms and characteristics (polymorphism)
- Can be combined with other molecules to create complex formulations and glasses
 - i.e. combine with aluminum oxide to create aluminosilicate glasses, used in GI cements

Formation of glasses: glass transitions

- Crystal structures follow a volume:temperature curve depicting a steep change in volume at a melting temperature
- Amorphous structures (glasses) do not have a discrete solid-liquid transition; the rate of change of the specific volume increases (change in slope)
 - Volume gradually increases/decreases along the curve, not suddenly
 - In cooling, the liquid will have large amount of unoccupied volume



Formation of glasses

- Glasses are formed when a liquid-form of a compound experiences gradual contraction / volume reduction, prohibiting orderly crystal formation and creating an amorphous structure
 - Larger, complex molecules make regular, orderly arrangements difficult to achieve --> higher probability of glass formation
- Below glass transition temperature, molecular movement is very viscous --> no more configurational change, structure is set
- Vitrification – the forming of a glass by quick cooling
 - Quick cooling of a highly viscous liquid does not allow time for molecules to achieve orderly arrangement quickly

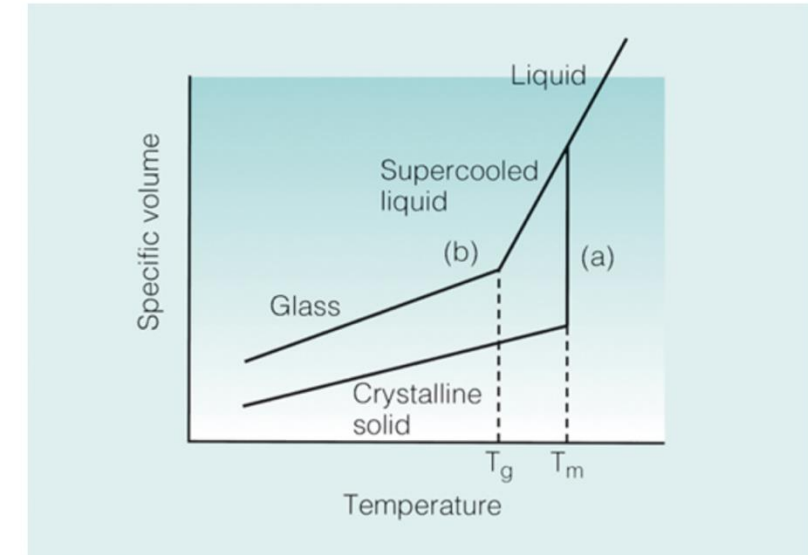
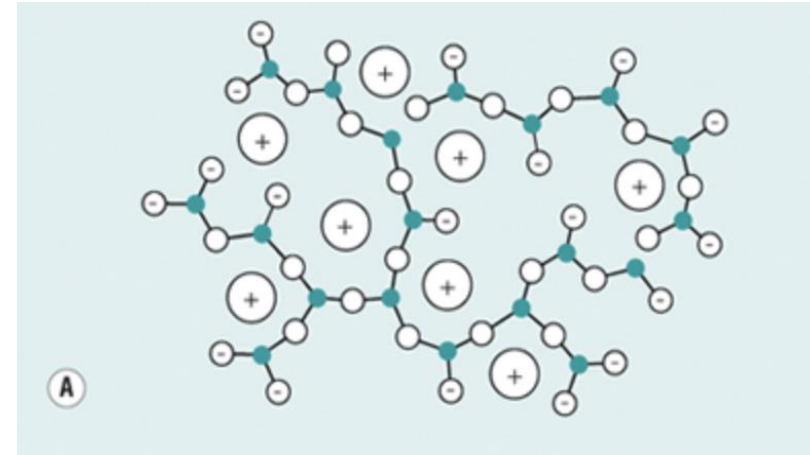
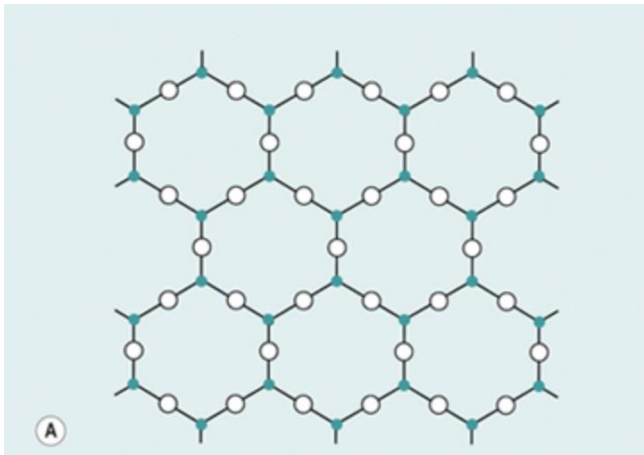


FIGURE 1.3.5 Cooling curves for a material that can form a crystalline solid (a) or a glass (b).

Glass formation in materials

- Silica is the foundation for dental glasses (glass former), but melting temperatures are too high for commercial use
- Use metal oxides (glass modifiers) to help reduce the melting temperature by interrupting silica's structure



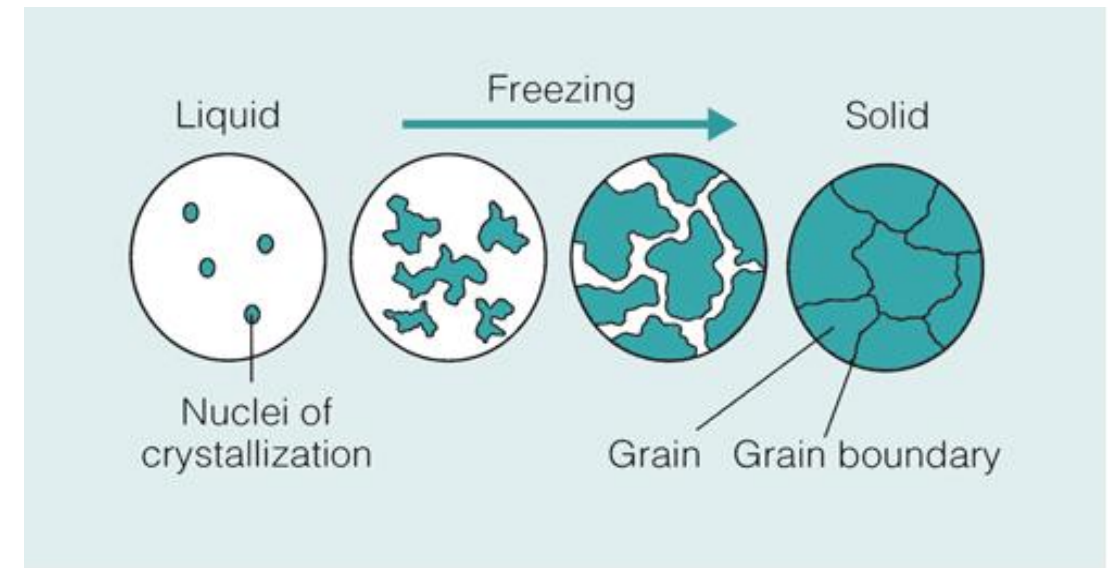
Formation of glass ceramics

- Devitrification - the crystallization of a glass
 - Occurs when glass is kept at an elevated temperature for more time
 - Allows more time for molecules to reassemble
- Achieves translucent appearance from scattering of light off surfaces of the newly formed small crystals



Structure of metals and alloys

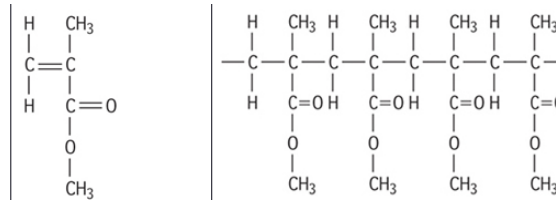
- Metals consist of crystalline nuclei formed when metals solidify; this forms *grains* – *analyzing them helps understand and manipulate metals*
- Alloy – mixture of 2+ metals or non-metallic elements
 - Consists of *phases* - structurally homogenous part of a system, separated by others physically
 - Many favorable properties, i.e. Thermal conductivity
 - May be brittle due to nonidentical distribution of metals throughout the mixture
 - In amalgams: Ag-Sn, Ag-Hg, Cu-Sn



Structure of polymers

- Polymers are long chain molecules, consisting of repeated monomers fused by polymerization

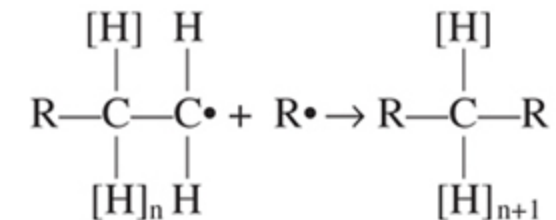
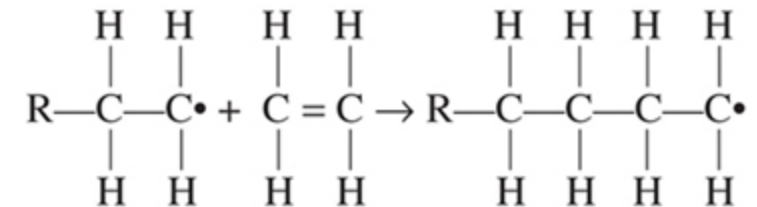
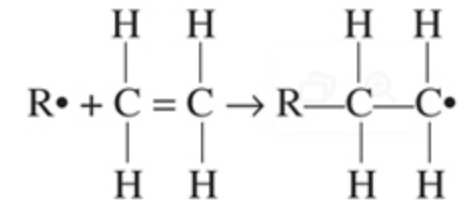
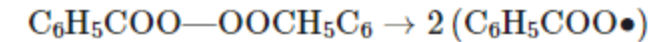
Polymethylmethacrylate



- The length of the chains is purely random; many different chain lengths may exist in a compound
- The polymer's molecular weight (# of repeating units * MW of unit), averaged out, determines physical properties
 - Higher average MW polymer requires stronger forces to be broken, but increases viscosity

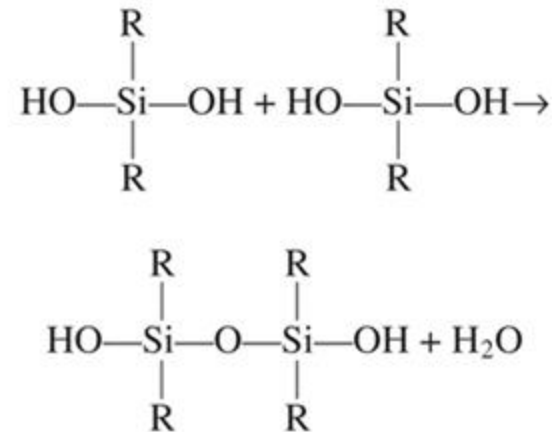
Addition polymerization

- Reaction between 2 molecules without the removal of a smaller molecule
- Occurs in 4 steps
 - Activation: by heat, chemical compounds, or light to produce free radicals (unpaired electrons)
 - i.e. camphorquinone in composites, benzoyl peroxide in denture resin
 - Initiation: free radical reacts with monomer, initiating polymerization
 - Propagation: free radical is transferred to monomer, which reacts with another monomer
 - Termination: free radicals react with one another to cease reaction



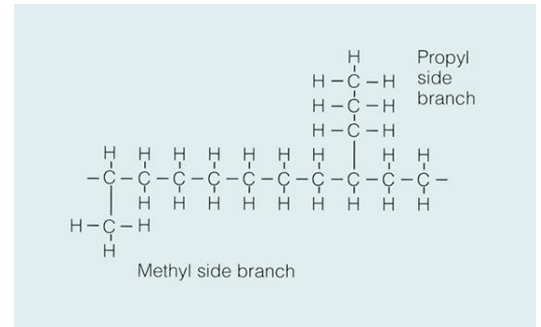
Condensation polymerization

- Any 2 molecules react to form a larger molecule, usually eliminating a smaller molecule (water)



Polymer chain configurations

- Glass transition temperature - temperature at which plastic softens, and polymer chains begin to flow
 - A plastic that can be softened and shaped with higher temperature is *thermoplastic*
- Linear chains flow easily, and are thus manipulated easily
- Branched chains have a 3D formation, preventing easy flow, thus maintaining overall polymer properties until extremely high temps



Cross-linking of polymers

- Polymer chains join together by chemical bonds are cross-linked
- Silicone polymers has a glass transition temperature lower than room temperature
 - liquid at room temp since chains can slide past each other
 - cross linking forms a solid at room temp, since chains can no longer slide past each other
- Extensive cross-linking: hard, brittle solids
- Sparse cross-linking: solid with some flexibility (returns to prior shape after some deformation)
 - Cross-linking can form complex coils of the polymer chains. These coils can be stretched (uncoiled) under stress, then recoil without stress
 - Amount of stretch determined by length of chains, amount of cross-linking, and strength of bonds

Elastomers

- Polymeric materials that can stretch by large amounts
 - Usually soft, with low elastic modulus (explained later)
 - Very high strains possible (explained later)
 - Strains are reversible (can recoil)
 - Material is easily manipulated and fluid at room temperature



Composites

- Combination of materials, in which individual components maintain identity, but synergistically produce favorable qualities
- Made of a *matrix* and *filler*
 - Matrices bind the filler together
 - Filler is usually introduced into a polymeric matrix to produce some function (color, strength, etc.)

